

Worksheet · Section 2.3–2.4

Extracting Information & Using Programs with Data



The Big Picture

Data becomes information when we find patterns in it. Practice telling data from metadata, cleaning messy data, filtering and sorting a table, and avoiding the correlation-causation trap.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|-------------------------------------|---|
| _____ 1. Data | A. Two variables change together |
| _____ 2. Information | B. A chart or graph that reveals patterns |
| _____ 3. Metadata | C. Data about data, like a photo's date |
| _____ 4. Data cleaning | D. A system's ability to handle growing amounts of data or users |
| _____ 5. Correlation | E. Changing every value in the same way, like °F to °C |
| _____ 6. Causation | F. Keeping only the rows that meet a condition |
| _____ 7. Filtering | G. Data that doesn't fairly represent the whole population |
| _____ 8. Sorting | H. Facts and patterns extracted from data |
| _____ 9. Transforming | I. Raw values, like numbers, text, or images |
| _____ 10. Data visualization | J. Arranging data in order |
| _____ 11. Scalability | K. Making data uniform without changing its meaning |
| _____ 12. Bias in data | L. One thing directly makes another happen |

Part 2 · Fill It In

Use each word from the word bank once.

visualization**causation****filter****metadata****correlation****bias****sort****cleaning**

1. The date and GPS location stored with a photo are _____.
2. Changing "N.Y." and "new york" to "New York" is data _____.
3. Two variables that rise together show a _____.
4. A correlation alone doesn't prove _____.
5. To keep only rows where grade = 12, you _____ the data.
6. To put scores from highest to lowest, you _____ them.
7. A scatter plot is a data _____.
8. Surveying only robotics-club members about school technology could introduce _____.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · Data or Metadata?

Check whether each item is the data itself or data about it.

Item	Data	Metadata
1. The pixels of a photo		
2. The date a photo was taken		
3. The body of an email		
4. The sender's email address		
5. A song's audio		
6. A song's length		
7. A document's file size		
8. The words in a document		

Part 4 · Clean It Up

A survey's "Grade" column contains: 9, 9th, Freshman, 10, tenth, (blank), 11, 12th

How many entries mean 9th grade? _____

What should happen to the blank entry? _____

Write one cleaning rule: _____

Name: _____

Class: _____

Date: _____

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Apply It

Part 5 · Work the Table

Title	Genre	Year	Rating	Minutes
Star Run	Action	2019	7.8	128
Lake House	Drama	2016	8.1	112
Laugh Lab	Comedy	2021	6.9	95
Iron Peak	Action	2022	8.4	141
Tiny Paws	Comedy	2018	7.2	88
Quiet Hour	Drama	2023	7.5	104
Max Out	Action	2017	6.5	119
Joke Box	Comedy	2023	8.0	101

a) Filter Genre = Action → how many rows?

_____ b) Highest-rated comedy?

c) Shortest movie released after 2017?

_____ d) Average rating of dramas?

e) Steps to find the longest movie released in 2020 or later: _____

Part 6 · Correlation or Causation?

A study finds that students who eat breakfast have higher test scores.

a) Correlation or causation? b) Name a possible third factor. c) How could researchers test whether breakfast causes higher scores?



Think:

Before trusting a pattern, ask: Who is missing from this data? Could a third factor explain it? Does the data even contain what I need?

Name: _____

Class: _____

Date: _____

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. Which is metadata for a video?

- A. The video's length
- B. The scenes in the video
- C. The music in the video
- D. The people in the video

★★ On Level

2. Which chart best compares the average scores of 5 different classes?

- A. Scatter plot
- B. Pie chart
- C. Line graph
- D. Bar graph

★★ On Level

3. Umbrella sales and car accidents both rise on certain days. What is the best explanation?

- A. Rainy weather may cause both
- B. Accidents cause umbrella sales
- C. The data must be wrong
- D. Umbrellas cause accidents

★★★ Exam Hard

4. A data set has 50 million rows and is too slow to analyze on one computer. What is the best solution?

- A. Use lossy compression
- B. Analyze it by hand
- C. Delete half the rows
- D. Use parallel or distributed processing

★★★ Exam Hard

5. A menu data set has: dish, price, and isVegan. Which TWO steps are needed to find the average price of vegan dishes? (*Select two.*)

- A. Sort by dish name
- B. Filter rows where isVegan = true
- C. Delete the price column
- D. Average the price column of the remaining rows

Part 8 · Written Response

You have data on students' daily screen time and grades. Describe two questions you could investigate, the processing you'd do (filter, sort, visualize), and one limitation of your conclusions.
